Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/20/2005
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Project Title:	Presidential Award - Processing Motion Signals in Complex Environments - Phase II		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
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Space Biology Special Category:	None		
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Zip Code:	94115	**Congressional District:**	8
Comments:			
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	An accurate representation of scene layout and of moving objects in the scene is required to navigate through complex environments. In the first part of the grant, our efforts were largely directed toward understanding how humans detect objects moving along a smooth path in a cluttered environment. More recently we have also looked at how depth information contributes to surface layout. . Our goal is to understand how local disparity and motion signals are combined to guide human locomotion and to provide information about depth, object boundaries and surface structure. Understanding how motion and depth information are combined in situations that include friction and gravity will help predict human perception under diverse conditions critical to NASA's mission of space exploration.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Earth Benefits: An accurate representation of scene layout and of moving objects in the scene is required to navigate through complex environments. Until recently our efforts were largely directed toward understanding how humans detect objects moving along a smooth path in a cluttered environment. Over the last year we have also begun to look at surfaces defined solely by disparity or depth information. Our goal is to understand how local disparity and motion signals are combined to guide human locomotion and to provide information about depth, object boundaries and surface structure.
Task Progress:	Segmentation by motion cues. This study investigated human ability to isolate a target based on different kinds of visual cues. Previously we had shown that the ability to discriminate changes in speed on an object of interest critically depended on how a stimulus was segmented (Verghese & Stone, 1996). When stimuli were seen as separate entities, independent information was available from each entity. But when these separate stimuli were grouped into a single patch, only a single estimate was available. In this study we determined whether a cue would overcome the effect of grouping and allow observers to select a part of a grouped stimulus. Two stimulus configurations were presented at an eccentricity of 3 degrees: the ungrouped stimulus was made up of 4 widely separated Gabor patches, and the grouped stimulus was a large Gabor divided into 4 quadrants. All stimuli had a spatial frequency of 1.6 c/deg and drifted at 2.4 deg/s. The observer had to choose which of two intervals had a region (patch or quadrant) with a contrast increment. The potential test region was cued with a clearly visible cue in each interval. We used three different cues: a high contrast spatial cue, a speed cue where the target region drifted at 4X times the speed of the other regions, and a direction cue where the target drifted in the opposite direction. The data across 4 observers showed a consistent pattern. When the patches were widely separated, cueing improved thresholds significantly over the uncued case, indicating that the cue unambiguously signaled the target location. Furthermore, in the presence of a cue, thresholds improved exactly as predicted from a perfect knowledge (no uncertainty) about the target. On the other hand, cueing one quadrant of the grouped stimulus barely improved contrast discrimination over the uncued case. The spatial cue had no effect at all, and the speed and direction cues were only slightly more effective. More importantly, thresholds with no cue in the grouped configuration were a factor of two higher than the cued case in the ungrouped configuration. This is larger than predicted from uncertainty about which one of 4 quadrants is the target: it suggests that observers are unable to segment the grouped stimulus and therefore sum the information from all 4 quadrants. The wallpaper illusion revisited. Shifts in the apparent depth of a repetitive pattern or texture occur when the eyes are stably converged at a position in front of the pattern. Observations on this phenomenon, known as the Wallpaper illusion, have been made for almost three centuries. Here, Suzanne McKee, and I examined whether the ability to discriminate small changes in depth follows the shifts in the apparent depth of the pattern. The repetitive pattern for our study was a vertical sinusoid with a spatial frequency of 3 cycles per degree, and with a horizontal and vertical subtense of 6 degrees and 2 degrees, respectively. As we have previously reported, this segment is initially matched at the disparity of the 'edges', meaning the starting and ending positions of the segment. However, if the observer converges for several seconds on a fixation point presented in front of and 30' below the segment, the segment shifts forward in depth by an even multiple of the period. For example, if the edges specify a depth 3/4 of a period behind the grating, the segment shifts forward to a disparity 1/4 of a period. Observers signaled when this shift occurred; at that point, we extinguished the fixation point, and 500 msec later presented a test probe at various disparities. The observer judged the relative disparity of the test with respect to the grating. Stereoacuity was always best in the plane corresponding to the perceived depth of the segment, whether at the 'edges' or at the new forward match. Thus, without any change in the stimulus, stereo sensitivity underwent dramatic changes triggered by perceptual shifts to alternative matches. An information maximization model of eye movements. We proposed a sequential information maximization model as a general strategy for programming eye movements. The model reconstructs high-resolution visual information from a sequence of fixations, taking into account the fall-off in resolution from the fovea to the periphery. From this framework we obtain a simple rule for predicting fixation sequences: after each fixation, fixate next at the location that minimizes uncertainty (maximizes information) about the stimulus. We obtained human eye movement data from a separated study where humans were presented with novel shapes that they viewed peripherally for 300ms, then actively studied for 1.2 seconds while eye movements are measured with an SRI Dual Purkinje Image eye tracker. After the learning interval, the studied shape reappears at a different size and location with a highly similar shape. The observers' task was to select the shape they saw and rate their confidence. WE compared these eye movement data to the predictions of our model as well as to another models that chooses "salient" locations or random locations as fixation targets, we demonstrate that our model is best at predicting fixation locations. Modeling additional biological constraints will improve the prediction of fixation sequences. Our results suggest that information maximization is a useful principle for programming eye movements.
Bibliography Type:	Description: (Last Updated: 03/15/2017)
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